Bank Marketing Success Prediction: A State-of-the-Art Machine Learning Approach

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Introduction – Motivation

Direct marketing remains a cornerstone for banking institutions to promote term deposits. However, unsolicited phone calls incur significant operational costs and risk customer dissatisfaction. The motivation of this project is to develop a highly accurate predictive model that can identify customers with a high probability of conversion and to be compared to the ones in existing literature (Moro et al.). By optimizing the targeting process, the bank can maximize successful deposits while drastically reducing the number of unnecessary contacts. Also from a machine learning perspective we get an analytical view on how enormously the performance of every model is affected, by the hyperparameter tuning and overall how even simple models like the Logistic Regression can surpass Neural networks if we conduct the appropriate tuning.

Problem Definition and Data Description

The problem is defined as a binary classification task: predicting whether a client will subscribe to a term deposit.

The study utilizes the "Bank Marketing" dataset (Moro et al.), containing 41,188 records. The features include:

- **Demographics:** Age, job, marital status, education.
- **Financial Indicators:** Default status, housing, and personal loans.
- **Socio-economic context:** Euribor 3m rate, consumer confidence index, and employment variation rate.
- **Campaign Data:** Number of contacts performed during the current and previous campaigns.

The data face a severe class imbalance where our target (y) consists of 89% class 0 (no) and 11% class 1 (yes), which is translated into 'no-buy' and 'buy' for the bank. In the following illustrations we will visualize data histograms, where we receive important information of their type and behaviour (e.g categorical, numerical, correlated, missing values, etc)

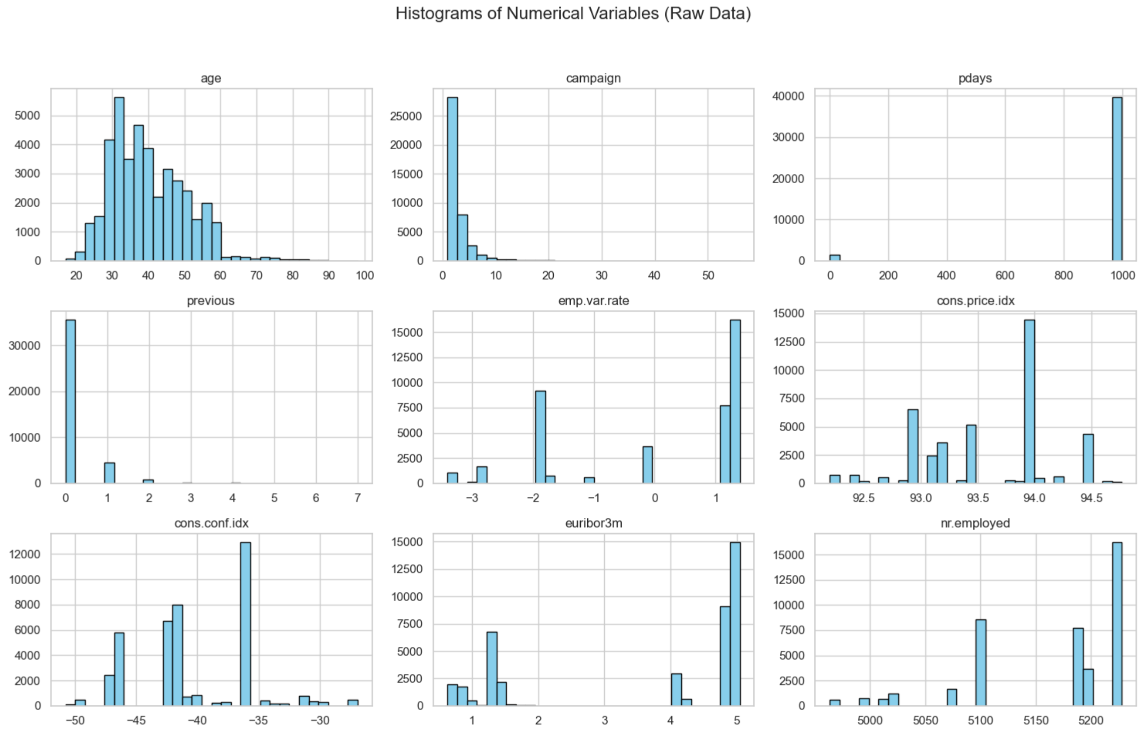

Figure 1: Numerical variable histograms

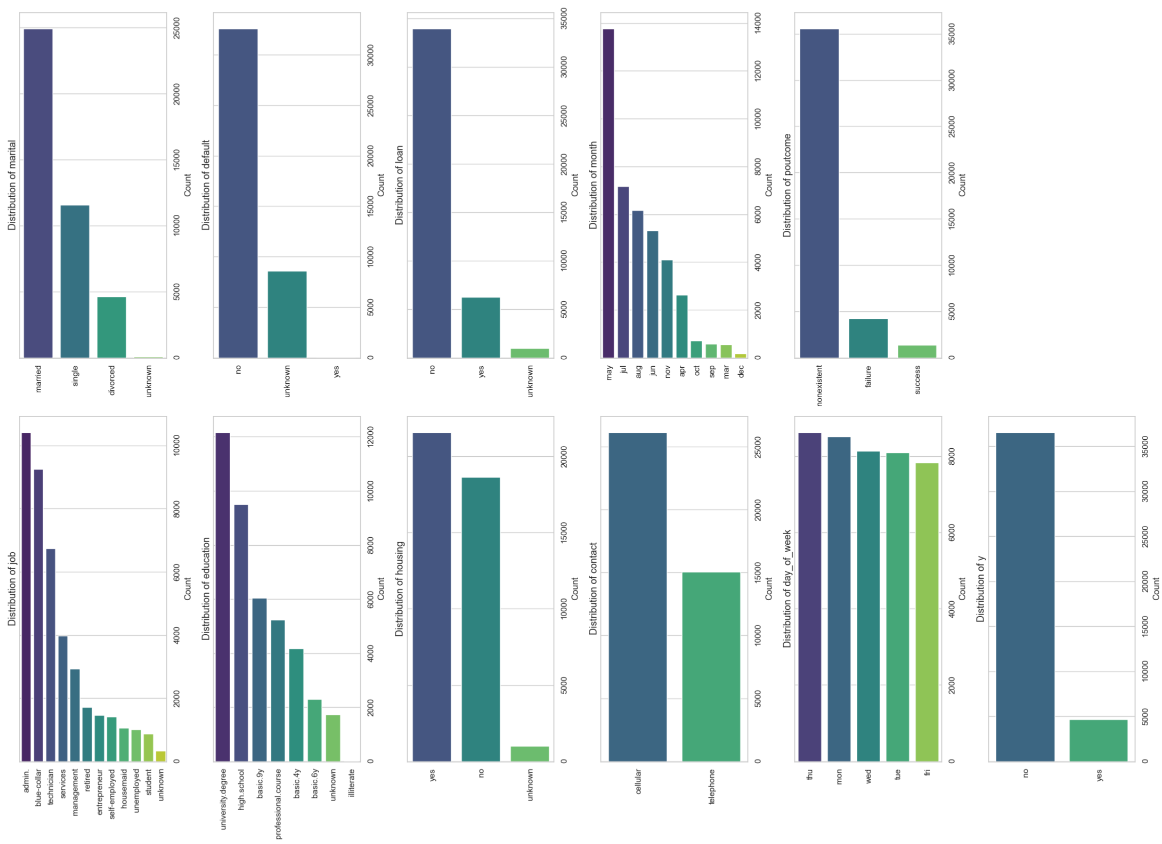

Figure 2: Categorical variable histograms

Solution Approach

0.1 Methods and Algorithms

A multi-algorithmic framework was established to evaluate a wide spectrum of predictive behaviors, ranging from traditional statistical methods to state-of-the-art ensemble learning. The following models were rigorously optimized using **GridSearchCV** combined with **5-fold Stratified Cross-Validation** to address the class imbalance:

- **Linear & Instance-based Baselines:**
  - **Logistic Regression:** Employed as the fundamental linear baseline to establish a performance floor.
  - **Support Vector Machines (SVM):** Utilized to capture potential non-linear decision boundaries through high-dimensional kernel mapping. For the SVM we used a smaller version of the original dataset provided from the authors, due to computational overload.
  - **K-Nearest Neighbors (KNN):** A non-parametric approach used to evaluate the local density and similarity between client profiles.
- **Ensemble & Boosting Techniques:**
  - **Random Forest:** An ensemble of Bagged Decision Trees, implemented to reduce variance and handle complex feature interactions.
  - **CatBoost:** A state-of-the-art Gradient Boosting Decision Tree (GBDT) framework. It was selected as the primary contender due to its **Ordered Boosting** mechanism and its superior native handling of categorical variables, which effectively minimizes information loss during the encoding process.

0.2 Implementation and Choices

The implementation was conducted in Python 3.13 using the `scikit-learn` and `catboost` libraries. A critical methodological choice, aligning with paper's findings was the **exclusion of the 'duration' variable** to prevent data leakage, as this information is only available after a call is completed. This was shown clearly from the performance of the models, since they achieved higher training AUC scores but significantly lower test AUC scores. Data was processed using a `Pipeline` including `SimpleImputer` and `StandardScaler` for baseline models, while CatBoost utilized its native categorical handling. To conclude we had to also set the 'pdays' and 'previous' variables as categorical via the OneHotEncoder for the linear models, since for instance the value 999 of 'pdays' corresponds to 'no communication' instead of 999 days passed from last call and this would drastically affect the weights of the linear models and would be misleading.

0.3 Relation to State-of-the-Art

The benchmark for this study is the work of Moro et al. (2014), which achieved an AUC of 0.794 using Neural Networks. Our approach evolves this by using modern Gradient Boosting and rigorous 5-fold Stratified Cross-Validation.

Experiments

0.4 Claims and Research Questions

We claim that by using CatBoost with optimized hyperparameters, we can exceed the 0.794 AUC threshold and provide a higher Lift for the bank's marketing operations.

0.5 Detailed Description and Results

The models were evaluated using a 5-fold Stratified Cross-Validation on the training set and a final evaluation on a hold-out test set.

- **CatBoost CV Performance:** Standard Deviation between AUC score performance of the folds (± 0.0047). Which means our model is robust.
- **Final Test AUC:** 0.8157 (Surpassing the 0.794 benchmark).

0.6 Results Comparison

Following the methodology of Moro et al. (2014), we extended our evaluation beyond standard accuracy metrics by calculating Cumulative Gains and Lift. These metrics are crucial for quantifying the 'marketing ROI' and demonstrating how the model optimizes the bank's contact strategy

Table 1: Business Impact: Cumulative Gains and Lift Comparison

Contacted Sample	Paper Recall	CatBoost Recall	Lift (CatBoost)	Difference
Top 10% (High Priority)	19.2%	**48.28%**	**4.83**	+29.08%
Top 20%	35.6%	**67.46%**	3.37	+31.86%
Top 50% (Mid Campaign)	78.9%	**84.27%**	1.69	**+5.37%**

The Lift Score of 4.83 at the top 10% of the population implies that the bank can capture nearly half (48.28%) of all total deposits by contacting only 10% of its client base.

Also we provide the effectiveness of our models, demonstrated through their AUC performance compared to the paper's benchmark:

Table 2: Comparative Performance of Evaluated Models vs. Paper Benchmark

Model	Test AUC	Improvement (%)	Status
CatBoost	**0.8157**	**+2.73%**	**Winner**
Random Forest (RF)	0.8138	+2.49%	Optimized
Logistic Regression (LR)	0.8012	+0.91%	Baseline
NN Paper Benchmark (2014)	*0.7940*	*0.00%*	*Reference*
KNN	0.7799	-1.78%	Baseline
SVM (Small Dataset)	0.7603	-4.24%	Limited

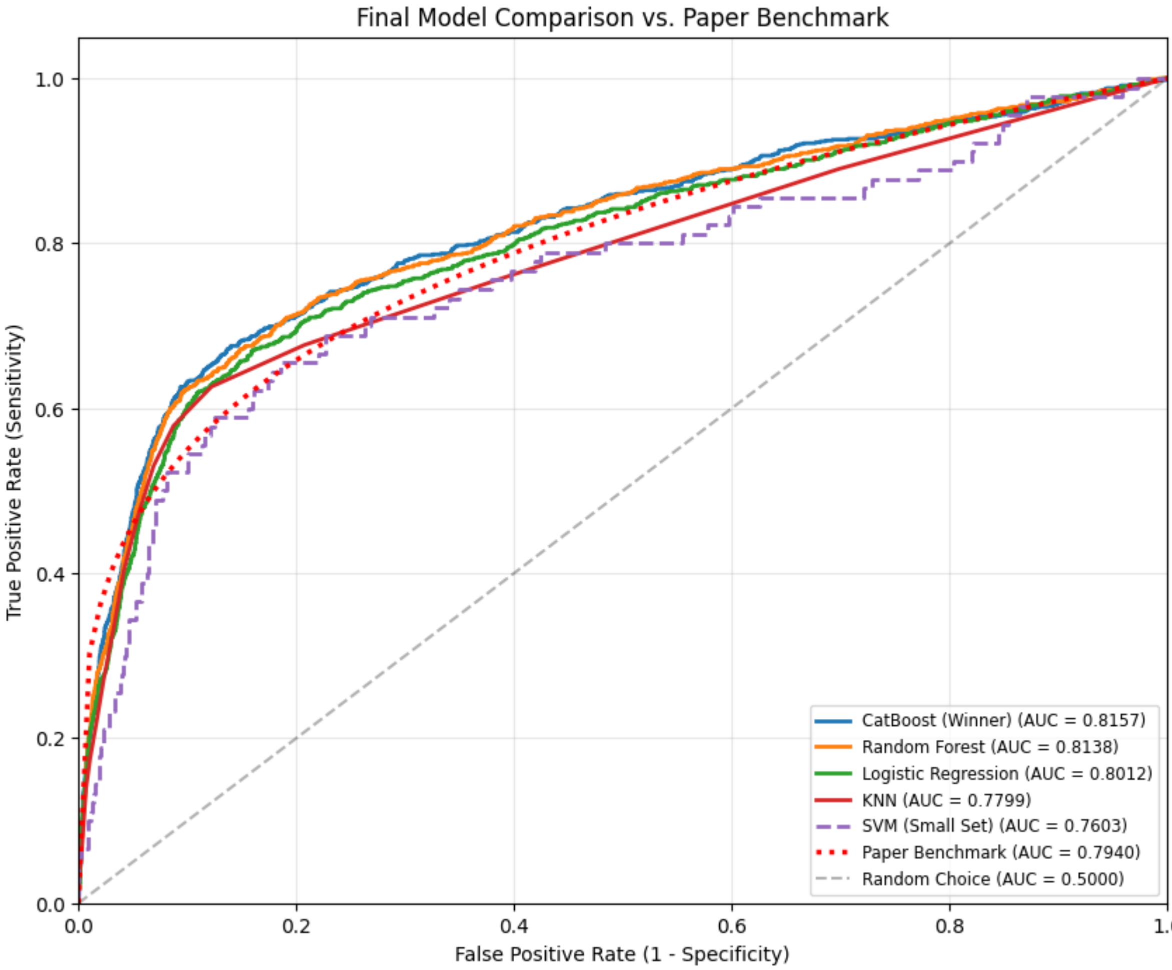

Figure 3: AUC Performance Comparison

0.7 Feature Importance

The interpretability of the model was assessed by analyzing the feature importance scores from the highest-performing classifier, in our case the CatBoost. The results highlight the following key drivers:

- **Euribor 3m Rate:** This socio-economic indicator emerged as the most significant feature, suggesting that term deposit subscriptions are highly sensitive to the prevailing interest rate environment.
- **Education:** Academic background serves as a strong demographic proxy for financial literacy and investment propensity.
- **Campaign:** Interestingly, the number of contacts performed during the current marketing campaign emerged as the third most critical feature. This suggests that the intensity of the bank's outreach strategy is a decisive factor in predicting the client's final response.

This ranking confirms that successful bank marketing is a combination of *timing* (macro-economics), *targeting* (demographics), and *engagement intensity* (current campaign outreach).

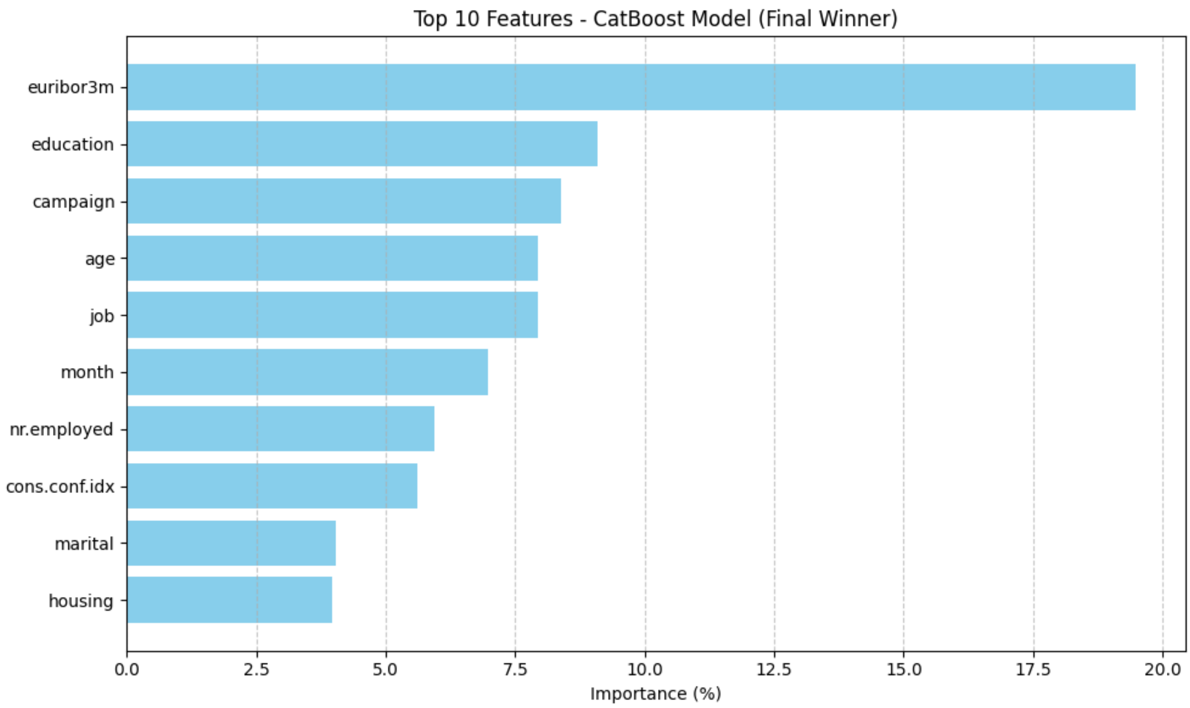

Figure 4: Top 10 features according to CatBoost

Conclusions – Discussion

The experimental results confirm that the CatBoost model is the superior choice for this task, significantly outperforming the Neural Network from the original paper. Additionally, with the appropriate hyperparameter tuning, we observed that even baseline models such as Logistic Regression can deliver astonishing results, using fewer computational resources compared to a Neural Network. The analysis of feature importance revealed that socio-economic indicators, specifically the **Euribor 3m rate**, are the most critical predictors.

In conclusion, our model provides a robust tool for banking institutions to optimize their marketing campaigns, ensuring higher conversion rates with significantly fewer resources. Future work could involve the use of Nested Cross-Validation to further validate model stability on smaller subsets of the data.